

Chapter 4

Functors and Natural Transformations

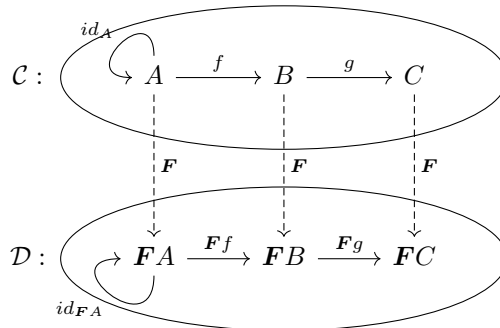
4.1 Functors

Definition 4.1.1 (Functor). A *functor* F between categories $\mathcal{C}$ and $\mathcal{D}$ (notation $F : \mathcal{C} \longrightarrow \mathcal{D}$) is specified by:

- an operation assigning to objects A in $\mathcal{C}$ to objects FA in $\mathcal{D}$
- an operation assigning to morphisms $f : A \longrightarrow B$ in $\mathcal{C}$ to morphisms $Ff : FA \longrightarrow FB$ in $\mathcal{D}$ which satisfies the following:

$$\begin{aligned} F(id_A) &= id_{FA} \\ F(g \circ f) &= Fg \circ Ff \end{aligned}$$

i.e. the following diagram commutes.



Remark 4.1.2. By a functor $F : \mathcal{A} \longrightarrow \mathcal{B}$, any commutative diagram in $\mathcal{A}$ is sent to a commutative diagram in $\mathcal{B}$.

Example 4.1.3 (Powerset functor). The *powerset functor* $\mathcal{P}ow(_) : \mathcal{S}et \longrightarrow \mathcal{S}et$ sends each object X to $\mathcal{P}ow(X)$ and each arrow $f : X \longrightarrow Y$ to $X \mapsto \{f(x) \mid x \in X\}$.

Example 4.1.4 (Product functor). Suppose that a category $\mathcal{C}$ has binary products. The *product functor* $_ \times A : \mathcal{C} \longrightarrow \mathcal{C}$ sends each objects B to $B \times A$ and each arrow $f : B \longrightarrow C$ to $f \times id_A$.

Example 4.1.5 (Exponentiation functor). Suppose that a category $\mathcal{C}$ is Cartesian closed. The *exponentiation functor* $_{}^A : \mathcal{C} \longrightarrow \mathcal{C}$ sends each object B to B^A and each arrow $f : B \longrightarrow C$ to $f^A : B^A \longrightarrow C^A$.

Example 4.1.6 (Homomorphism). The *homomorphisms* between posets are functors.

Exercise 4.1.7. Confirm that each of the examples above actually conforms to the definition of functor.

Example 4.1.8. The functor $F : \mathcal{S}et \longrightarrow \mathcal{M}on$ takes a set to its Kleene closure, which sends each set A to A^* , and each set function f to $map(f)$, where $map(f) : A^* \longrightarrow B^*$ is defined by:

$$map(f)([a_1, \dots, a_n]) \stackrel{def}{=} [f(a_1), \dots, f(a_n)]$$

with $[a_1, \dots, a_n]$ any elements of A^* . To verify that F is indeed a functor, check:

$$\begin{aligned} F(id_A)([a_1, \dots, a_n]) &\stackrel{def}{=} map(id_A)([a_1, \dots, a_n]) \\ &= id_{A^*}([a_1, \dots, a_n]) \\ &\stackrel{def}{=} id_{F A}([a_1, \dots, a_n]) \end{aligned}$$

$$\begin{aligned} F(gf)([a_1, \dots, a_n]) &\stackrel{def}{=} map(gf)([a_1, \dots, a_n]) \\ &= [gf(a_1), \dots, gf(a_n)] \\ &= map(g)([f(a_1), \dots, f(a_n)]) \\ &= map(g)(map(f)([a_1, \dots, a_n])) \\ &= Fg \circ Ff([a_1, \dots, a_n]) \end{aligned}$$

Exercise 4.1.9. *Cat* is a category of categories with categories as objects and functors as arrows.

1. For each category $\mathcal{A}$, define a functor $id_{\mathcal{A}}$ as an identity functor for $\mathcal{A}$.
2. Show that functors are composable and the composition is unitary with the identity functor above, and associative.

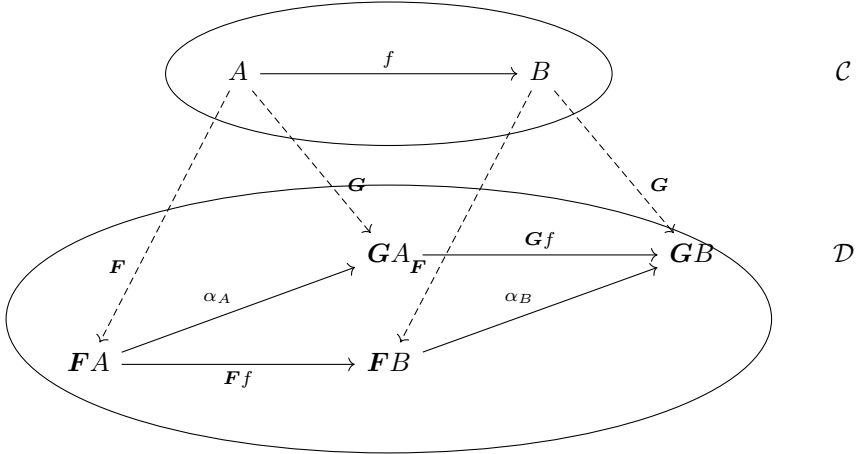
4.2 Natural Transformations

Definition 4.2.1 (Natural Transformations). Let $\mathcal{C}$ and $\mathcal{D}$ be categories and $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be functors. A *natural transformation* α from F to G (notation $\alpha : F \rightrightarrows G$) is specified by:

- an operation assigning to each object A in $\mathcal{C}$ a morphism $\alpha_A : FA \rightarrow GA$ in $\mathcal{D}$, such that for any morphism $f : A \rightarrow B$ in $\mathcal{C}$, we have:

$$Gf \circ \alpha_A = \alpha_B \circ Ff$$

i.e. the following diagram commutes.



Example 4.2.2. Given the functor $F : Set \rightarrow Mon$, we can define a natural transformation $rev : F \rightrightarrows F$ which has components $rev_A : A^* \rightarrow A^*$ defined

by:

$$rev_A([a_1, \dots, a_n]) \stackrel{def}{=} [a_n, \dots, a_1]$$

where A is a set and $[a_1, \dots, a_n] \in A^*$. To verify that this is indeed a natural transformation, check:

$$\begin{aligned} Ff \circ rev_A([a_1, \dots, a_n]) &= [f(a_n), \dots, f(a_1)] \\ &= rev_B \circ Ff([a_1, \dots, a_n]) \end{aligned}$$

Exercise 4.2.3. Suppose that $F, G : \mathcal{D} \rightarrow \mathcal{E}$, $I : \mathcal{C} \rightarrow \mathcal{D}$, $J : \mathcal{E} \rightarrow \mathcal{F}$ are functors and $\alpha : F \rightarrowtail G$ is a natural transformation. Verify that:

1. there is a natural transformation $\alpha_I : FI \rightarrowtail GI$ with components $(\alpha_I)_C \stackrel{def}{=} \alpha_{IC}$ where $C \in ob(\mathcal{C})$.
2. there is a natural transformation $J\alpha : JF \rightarrowtail JG$ with components $(J\alpha)_D \stackrel{def}{=} J(\alpha_D)$ where $D \in ob(\mathcal{D})$.

Exercise 4.2.4. Suppose that $\mathcal{C}$ is Cartesian closed and A is an object of $\mathcal{C}$. Verify that $(_{}^A) \times A \rightarrowtail id_{\mathcal{C}}$ is a natural transformation with components $ev_A : (_{}^A) \times A \rightarrow id_{\mathcal{C}}$.

Definition 4.2.5 (Natural Isomorphism). A natural transformation with every component iso is called a *natural isomorphism*. If there is a natural isomorphism between the functors F and G , then F and G are *naturally isomorphic*.

Example 4.2.6. If a category $\mathcal{A}$ has binary products, then $_{} \times 1 \rightarrowtail id_{\mathcal{A}}$ is a natural isomorphism.

4.3 Functor Categories

Definition 4.3.1 (Functor Category). A *functor category* of $\mathcal{C}$ and $\mathcal{D}$ (notation $[\mathcal{C}, \mathcal{D}]$ or $\mathcal{D}^{\mathcal{C}}$) is a category with objects functors from $\mathcal{C}$ to $\mathcal{D}$, morphisms natural transformation between such functors.

Exercise 4.3.2. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, $\mathbf{F}, \mathbf{G}, \mathbf{H}$ be functors from $\mathcal{C}$ to $\mathcal{D}$, and $\alpha : \mathbf{F} \xrightarrow{\sim} \mathbf{G}$ and $\beta : \mathbf{G} \xrightarrow{\sim} \mathbf{H}$ be natural transformations.

We can define a natural transformation $\beta \circ \alpha : \mathbf{F} \xrightarrow{\sim} \mathbf{H}$ by setting its components to be $(\beta \circ \alpha)_A \stackrel{\text{def}}{=} \beta_A \circ \alpha_A$. Verify the following:

1. the natural transformation defined above is indeed a natural transformation.
2. functor categories are indeed categories.

Exercise 4.3.3. Show that a natural isomorphism is an isomorphism in a functor category, by the following steps:

1. Given a natural isomorphism α in $\mathcal{D}^{\mathcal{C}}$ with inverse β , show that each β_C is an inverse for α_C in $\mathcal{D}$.
2. Given a natural transformation α in $\mathcal{D}^{\mathcal{C}}$ for which each component α_C has an inverse (say β_C) in $\mathcal{D}$, show that the β_C are the components of a natural transformation β which is the inverse of α in $\mathcal{D}^{\mathcal{C}}$.

Definition 4.3.4 (Product of categories). The *product* of $\mathcal{C}$ and $\mathcal{D}$ (notation $\mathcal{C} \times \mathcal{D}$) has objects pairs (C, D) where $C \in \text{ob}(\mathcal{C})$ and $D \in \text{ob}(\mathcal{D})$. The morphisms $(C, D) \rightarrow (C', D')$ are pairs of morphisms (f, g) where $f : C \rightarrow C'$ and $g : D \rightarrow D'$.

Exercise 4.3.5. We can define a functor $Ev : \text{Set}^{\mathcal{C}} \times \mathcal{C} \rightarrow \text{Set}$, which is defined as follows:

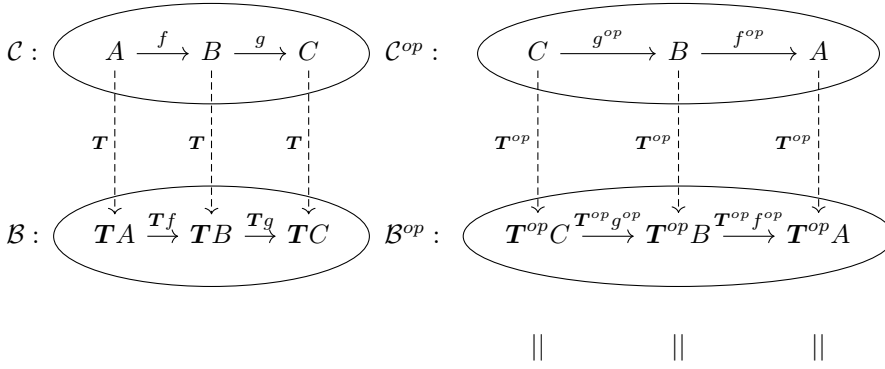
$$\begin{aligned} Ev(\mathbf{F}, A) &\stackrel{\text{def}}{=} \mathbf{F}A \\ Ev(\mu, g) &\stackrel{\text{def}}{=} \mu_{A'} \circ \mathbf{F}g = \mathbf{F}'g \circ \mu_A \end{aligned}$$

where $g : A \rightarrow A'$ and $\mu : \mathbf{F} \xrightarrow{\sim} \mathbf{F}'$. Show that Cat is Cartesian closed with functor categories as exponentials.

Definition 4.3.6 (Equivalence). Two categories $\mathcal{C}$ and $\mathcal{D}$ are *equivalent* (notation $\mathbf{F} : \mathcal{C} \simeq \mathcal{D} : \mathbf{G}$) if there are functors $\mathbf{F} : \mathcal{C} \rightarrow \mathcal{D}$ and $\mathbf{G} : \mathcal{D} \rightarrow \mathcal{C}$ together with natural isomorphisms $\varepsilon : \mathbf{F}\mathbf{G} \cong \text{id}_{\mathcal{D}}$ and $\eta : \text{id}_{\mathcal{C}} \cong \mathbf{G}\mathbf{F}$. We call such $\mathbf{F}$ and $\mathbf{G}$ *functor isomorphisms*.

4.4 Covariant and Contravariant Functors

Definition 4.4.1 (Dual Functor). If $T : \mathcal{C} \rightarrow \mathcal{B}$ is a functor, then the dual of T , written as $T^{op} : \mathcal{C}^{op} \rightarrow \mathcal{B}^{op}$ is also a functor, its object function $c \mapsto Tc$ and its mapping function $f^{op} \mapsto (Tf)^{op}$.

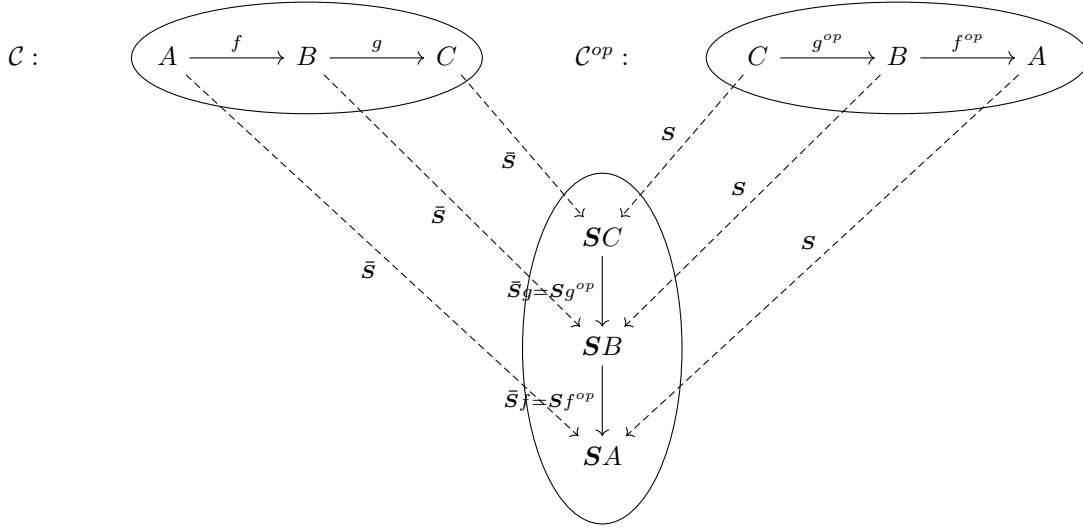


$$(TC)^{op} \xrightarrow{(Tg)^{op}} (TB)^{op} \xrightarrow{(Tf)^{op}} (TA)^{op}$$

Consider a functor $S : \mathcal{C}^{op} \rightarrow \mathcal{B}$ with the object function $A \mapsto SA$ and the mapping function $f^{op} \mapsto Sf^{op}$. This functor S may be expressed in terms of the original category $\mathcal{C}$ in the following way.

Definition 4.4.2 (Contravariant Functor). The *contravariant functor* $\bar{S} : \mathcal{C} \rightarrow \mathcal{B}$ of a functor $S : \mathcal{C}^{op} \rightarrow \mathcal{B}$ assigns each object A in $\mathcal{C}$ an object SA in $\mathcal{B}$ and each morphism $f : A \rightarrow B$ in $\mathcal{C}$ a morphism $Sf^{op} : SB \rightarrow SA$ in $\mathcal{B}$, in such a way that the following equations hold for any object A and any morphism f, g in $\mathcal{C}$.

$$\begin{aligned} \bar{S}(id_A) &= id_{\bar{S}A} \\ \bar{S}(g \circ f) &= \bar{S}f \circ \bar{S}g \end{aligned}$$



Remark 4.4.3. An ordinary functor defined as in Definition 4.1.1 is called a *covariant functor* in contradistinction to contravariant functors. It is often more convenient to represent a contravariant functor $\bar{S}$ on $\mathcal{C}$ to $\mathcal{B}$ as a covariant functor $S : \mathcal{C}^{op} \rightarrow \mathcal{B}$ or as a covariant functor $S^{op} : \mathcal{C} \rightarrow \mathcal{B}^{op}$.

Example 4.4.4. A *covariant powerset functor* $P : \mathcal{Set} \rightarrow \mathcal{Set}$ is given by

$$f : A \rightarrow B \mapsto f_* : PA \rightarrow PB$$

where $f : A \rightarrow B$ is a function and f_* is defined by $f_*(A') \stackrel{def}{=} \{f(a') \mid a' \in A'\}$ where $A' \in P(A)$.

Example 4.4.5. A *contravariant powerset functor* $P : \mathcal{S}^{op} \rightarrow \mathcal{Set}$ is given by

$$f : A \rightarrow B \mapsto f^{-1} : P(B) \rightarrow P(A)$$

where $f : A \rightarrow B$ is a function in $\mathcal{Set}$, and the function f^{-1} is defined by $f^{-1}(B') \stackrel{def}{=} \{a \in A \mid f(a) \in B'\}$ where $B' \in P(B)$.

4.5 Full and Faithful Functors

Definition 4.5.1 (Full Functor). A functor $F : \mathcal{A} \longrightarrow \mathcal{B}$ is *full* iff, for any objects A, B of $\mathcal{A}$ and any morphism $g : FA \longrightarrow FB$ of $\mathcal{B}$, there is some $f : A \longrightarrow B$ in $\mathcal{A}$ such that $Ff = g$.

Definition 4.5.2 (Faithful Functor). A functor $F : \mathcal{A} \longrightarrow \mathcal{B}$ is *faithful* iff, for any objects A, B and any morphisms $f, g : A \longrightarrow B$ of $\mathcal{A}$, if $Ff = Fg$ then $f = g$.

Remark 4.5.3. Clearly the composite of two full functors is a full functor and the composite of two faithful functors is a faithful functor.

Example 4.5.4. Given a pair of objects $A, B \in \mathcal{C}$, the arrow function of $T : \mathcal{C} \longrightarrow \mathcal{B}$ sends each arrow $f : A \longrightarrow B$ to $Tf : TA \longrightarrow TB$, so it defines a function:

$$T_{A,B} : \mathcal{C}(A, B) \longrightarrow \mathcal{B}(TA, TB), \quad f \mapsto Tf$$

Then T is full when every such function is surjective, and faithful when every such function is injective.

Remark 4.5.5. For a functor both full and faithful, every such function is a bijection. But this does not mean that the functor itself is an isomorphism of categories, for there may be objects of $\mathcal{B}$ not in the image of T .¹

¹MacLane (1997, p.15)

Bibliography

- Crole, R. L. (1993) *Categories for Types*. Cambridge University Press.
- Kleisli, H. (1965) “Every standard construction is induced by a pair of adjoint functors”, In the Proceedings of *the American Mathematical Society*, Vol. 16. pp.544–546.
- MacLane, S. (1997) *Categories for the Working Mathematician*, Graduate Texts in Mathematics. Springer, second edition edition.
- MacLane, S. and I. Moerdijk. (1992) *Sheaves in Geometry and Logic – A First Introduction to Topos Theory* – Springer.
- McLarty, C. (1992) *Elementary Categories, Elementary Toposes*, Oxford Logic Guides 21. Oxford Science Publications.
- Moggi, E. (1989) “Computational lambda-calculus and monads”, In the Proceedings of *Fourth Annual IEEE Symposium on Logic in Computer Science*. pp.14–23.
- Pedicchio, M. C. and W. Tholen. (2004) *Categorical Foundations: Special Topics in Order, Topology, Algebra, and Sheaf Theory*, Vol. 97 of *Encyclopedia of Mathematics and Its Applications*. Cambridge University Press.